CHITKARA UNIVERSITY

Course Handout

Institute/School Name	Chitkara University Institute of Engineering & Technology		
Department Name	Department of Computer Science & Engineering		
Programme Name	Bachelor of Engineering- Computer Science & Engineering **(Artificial Intelligence and Machine Learning)**		
Course Name	Back End Engineering	Session	July-Dec 2026
Course Code	24CAI0303	Semester/Batch	5th /2024
L-T-P (Per Week)	3-0-2	Course Credits	4
Pre-requisite	Software Engineering, Operating System, Computer Networks, DBMS	NHEQF Level	5.5
Course Coordinator	Dr. Palak Goyal	SDG Number	4,8,9

1. Objectives of the Course

The Back End Engineering course aims to equip students with the knowledge and practical skills required to design, develop, and maintain robust server-side applications for modern software systems. The course focuses on backend programming, database management, API development, authentication and authorization, server architecture, cloud integration, performance optimization, and deployment practices. Students will learn how to build secure, scalable, and maintainable backend services using industry-standard frameworks, relational and NoSQL databases, RESTful APIs, and version control systems. By the end of the course, students will be able to develop efficient backend solutions, integrate databases and external services, implement secure authentication mechanisms, optimize application performance, and deploy applications following modern software engineering principles.

The main objectives of the course are:

• To apply the concepts of web development to solve real life problems.

• To develop efficient solutions for logical problems using Node.js.

• Exercise and reinforce prior programming knowledge to effectively code standard problems.

• To identify and remove bugs in Backend code.

2. Course Learning Outcomes (CLOs)

Students should be able to:

	Course Learning Outcomes	POs	NHEQF Level Descriptor	No. of lectures
CLO01	Explain the fundamentals of backend engineering, including Internet architecture, client-server communication, HTTP/HTTPS protocols, Node.js architecture, and the role of backend servers in modern web applications.	PO1, PO2, PO11	Q2	9
CLO02	Develop backend applications using Node.js and Express.js by utilizing built-in modules, NPM packages, middleware, asynchronous programming concepts, debugging techniques, and RESTful API development.	PO1, PO3, PO4, PO5, PO10	Q3	21
CLO03	Design and integrate databases using MongoDB and Mongoose, perform CRUD operations, implement authentication and authorization using JWT, and develop secure backend services.	PO1, PO2, PO3, PO5	Q3	15

CLO04	Build scalable backend applications by implementing real-time communication with Socket.io, server-side templating, version control using Git and GitHub, API security, caching, and performance optimization techniques.	PO3, PO5, PO8, PO10, PO11	Q4	9
CLO05	Deploy and integrate AI models into backend applications using FastAPI/Flask, Docker, and Kubernetes while applying model serving, monitoring, versioning, and production deployment practices.	PO3, PO5, PO6, PO7, PO10, PO11	Q4	6
Total Contact Hours				**60**

CLO-PO-PSO Mapping grid | Program outcome (POs) and Program Specific Outcomes (PSOs) are available as a part of Academic Program Guide

Course Learning Outcomes	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**	**PSO3**	**Type of Assessment**
CLO01	H	L									H		L	H	Formative /Summative
CLO02	L		M	H	M					L		L		H	Formative /Summative
CLO03	M	H	H		H										Formative /Summative
CLO04			L		M			H		H	H				Formative /Summative
CLO05			M		H	M	L			M	H		M	H	Formative /Summative

*H=High, M=Medium, L=Low

3. Recommended Books

B01: “Web Development with Node and Express: Leveraging the JavaScript Stack” by Ethan Brown, 2019 **(**2nd Edition**)**
B02: “Node Cookbook: Discover Solutions, Techniques, and Best Practices for Server-Side Web Development with Node.js” by David Mark Clements, 2022 **(**4th Edition**)**
B03: “RESTful Web APIs: Services for a Changing World” by Leonard Richardson and Mike Amundsen, 2013 **(**1st Edition**)**
B04: “MongoDB: The Definitive Guide” by Kristina Chodorow, 2019 **(**3rd Edition**)**
B05: “JavaScript and jQuery: Interactive Front-End Web Development” by Jon Duckett, 2014 **(**1st Edition**)**
B06: “Practical MLOps: Operationalizing Machine Learning Models” by Noah Gift and Alfredo Deza, 2021 **(**1st Edition**)**

4. Other readings & relevant websites

Serial No	**Link of Journals, Magazines, websites and Research Papers**
1	http://www.w3schools.com/
2	http://www.javaScriptinfo.io/
3	https://nodejs.org/en
4	http://www.nptelvideos.com/javascript
5	https://www.freecodecamp.org/news/what-exactly-is-node-guide-for-beginners/

5. Recommended Tools and Platforms

Node.js, Express.js, MongoDB, MongoDB Compass, Visual Studio Code, NPM.Python, Jupyter Notebook, Google-Colab, Visual Studio Code, Anaconda.

6. Course Plan

Lecture Plan

Lecture Number	Topics
1-2	Introduction to JavaScript, Role of JavaScript in Web Development, Variables (var, let, const), Data Types
3-4	Type Conversion, Operators (Arithmetic, Assignment, Comparison, Logical), Conditional Statements
5-6	Loops (for, while, do-while, for...of, for...in), Practical Programming Exercises
7-8	Functions, Function Declaration, Function Expression, Parameters, Arguments, Return Values
9-10	Arrow Functions, Scope, Arrays, Creating and Accessing Arrays
11-12	Array Methods (push, pop, shift, unshift, splice, slice), Array Iteration
13-14	Higher Order Functions (forEach, map, filter, reduce, sort)
15-16	Objects, Properties, Methods, Nested Objects, Object Destructuring, JSON
17-18	DOM Introduction, Selecting Elements, querySelector(), querySelectorAll(), getElementById()
19-20	DOM Manipulation, Styles, Creating and Removing Elements, DOM Traversal
21	Event Handling (Mouse, Keyboard, Form Events), Event Object, preventDefault()
22	Forms, Form Validation, User Input Handling, Error Messages
23	Local Storage, Session Storage, Browser Data Management
24	ES6 Features (Template Literals, Spread, Rest, Destructuring, Default Parameters)
25-26	Callbacks, Asynchronous JavaScript, Promises, Promise Chaining, Error Handling
27-28	Async/Await, Fetch API, REST API Consumption, JSON Handling
29-30	JavaScript Mini Project with API Integration, Local Storage, Form Handling
Continuous Assessment (Lectures 1–30)	
31	Introduction to React, Why React, Single Page Applications, Vite Setup, Project Structure, JSX Fundamentals
32-33	Components, Functional Components, Component Reusability, Component Organization, Reusability
34	Props, Parent-to-Child Communication
35-36	State Management using useState(), Multiple State Variables
37	Event Handling in React, Forms, Controlled Components
38	Conditional Rendering, Lists and Keys
39-40	useEffect(), Side Effects, API Calls, Dependency Array
41	React Router, BrowserRouter, Routes, Navigation
42	Route Parameters, Nested Routes, Layout Components
43	Context API, useContext(), Global State Management
44	Axios, API Integration, Loading & Error States

45	Tailwind CSS Basics, Responsive Layouts
46	Authentication UI (Login, Registration, Validation)
47-48	React Project Development, Reusable Components, API Integration, Best Practices
49	Introduction to Node.js, History, Features, V8 Engine, REPL, Global Objects
50	Event Loop, Call Stack, Callback Queue, Non-Blocking I/O
51	Modules (CommonJS & ES Modules), File System Module (fs), Path Module
52	Buffers, Streams, EventEmitter, Error Handling
53	HTTP Module, Request-Response Lifecycle, Creating HTTP Server, Routing
54	NPM, package.json, Nodemon, Environment Variables
55	Introduction to Express.js, Express Server, Routing, Request & Response Objects
56	Middleware, Body Parsing, REST API Fundamentals, CRUD APIs
57	Validation, Error Handling Middleware, JWT Authentication, bcrypt, Protected Routes
58	Multer File Uploads, API Testing using Postman
59	MongoDB, MongoDB Atlas, CRUD Operations, Mongoose (Schemas, Models, Validation, Populate)
60	Connecting React with Express APIs, CORS, Authentication Flow, JWT Storage, Protected Routes, Final MERN Project Demonstration & Code Review
Final Assessment (Full Syllabus)	

Lab Plan

Lab No.	Experiment
1	Install and configure Visual Studio Code, Node.js, NPM, and Vite. Write basic JavaScript programs demonstrating variables, data types, operators, conditional statements, loops, and functions.
2	Develop JavaScript programs using arrays, objects, higher-order functions (map, filter, reduce, forEach), ES6 features, and JSON manipulation.
3	Create interactive web pages using DOM manipulation, event handling, form validation, Local Storage, and Session Storage.
4	Implement asynchronous JavaScript using Callbacks, Promises, Async/Await, Fetch API, and consume a public REST API.
5	Set up a React application using Vite and develop reusable functional components using JSX, Props, State (useState), and event handling.
6	Develop React applications implementing conditional rendering, lists, controlled forms, and useEffect() for API integration.
7	Build a React application using React Router, Context API, Axios, Tailwind CSS, and authentication user interface (Login and Registration Forms).
8	Develop a mini React project integrating REST APIs, reusable components, local storage, and form validation.
9	Install and configure Node.js. Develop programs using Node.js modules (Path, File System, Events), Buffers, Streams, and EventEmitter.
10	Create HTTP servers using the HTTP module, implement routing, query parameters, and serve HTML and JSON responses.

11	Develop Express.js applications implementing routing, middleware, request/response handling, body parsing, and RESTful CRUD APIs. Test APIs using Postman.
12	Implement input validation, centralized error handling, file upload using Multer, JWT authentication, password hashing (bcrypt), and protected routes in Express.js.
13	Install and configure MongoDB and MongoDB Atlas. Perform CRUD operations using MongoDB Compass and integrate MongoDB with Express using Mongoose Schemas and Models.
14	Integrate React frontend with Express backend and MongoDB database. Implement CORS, authentication flow, JWT storage, login/logout functionality, and protected routes.
15	Develop and demonstrate a complete MERN Stack project incorporating React, Express, Node.js, MongoDB, REST APIs, authentication, API integration, responsive UI, and CRUD operations following coding best practices.

7. Delivery/Instructional Resources

Plan (Theory +Lab):

Lec. No.	Topics	CLO	Book & Chapter	TLM	ALM	PPT URL	Web Reference	Audio-Video Link
1-2	Introduction to JavaScript, Role of JavaScript in Web Development, Variables, Data Types	CLO01	Book1 Ch1	PPT, Whiteboard	Think-Pair-Share	https://github.com/ssagar1999/chitkara_ppt	https://developer.mozilla.org/en-US/docs/Web/JavaScript	NPTEL – JavaScript
3-4	Type Conversion, Operators, Conditional Statements	CLO01	Book1 Ch2	PPT, Coding Demo	Hands-on Activity	https://github.com/ssagar1999/chitkara_ppt	https://javascript.info	NPTEL – JavaScript
5-6	Loops, Practical Programming Exercises	CLO01	Book1 Ch2	PPT, Live Coding	Problem Solving	https://github.com/ssagar1999/chitkara_ppt	https://developer.mozilla.org	JavaScript Tutorial
7-8	Functions, Parameters, Arguments, Return Values	CLO01	Book1 Ch3	PPT, Coding Demo	Pair Programming	https://github.com/ssagar1999/chitkara_ppt	https://javascript.info/function-basics	NPTEL – JavaScript
9-10	Arrow Functions, Arrays, Array Operations	CLO01	Book1 Ch4	PPT, IDE Demo	Coding Exercise	https://github.com/ssagar1999/chitkara_ppt	https://developer.mozilla.org	JavaScript Arrays
11-12	Array Methods, Higher Order Functions	CLO01	Book1 Ch4	PPT, Live Demo	Hands-on Practice	https://github.com/ssagar1999/chitkara_ppt	https://developer.mozilla.org	JavaScript Higher Order Functions
13-14	Objects, JSON, Object Destructuring	CLO01	Book1 Ch5	PPT, Coding Demo	Collaborative Learning	https://github.com/ssagar1999/chitkara_ppt	https://developer.mozilla.org	NPTEL – JavaScript

15-16	DOM Introduction, Element Selection, DOM Manipulation	CLO01	Book1 Ch6	PPT, Browser Demo	Lab Activity	https://github.com/ssagar1999/chitkara_ppt	https://developer.mozilla.org/docs/Web/API/Document_Object_Model	DOM Tutorials
17-18	Event Handling, Forms, Validation	CLO01	Book1 Ch7	PPT, Live Demo	Hands-on Activity	https://github.com/ssagar1999/chitkara_ppt	https://developer.mozilla.org	JavaScript Events
19-20	Local Storage, Session Storage, ES6 Features	CLO01	Book1 Ch8	PPT	Problem Solving	https://github.com/ssagar1999/chitkara_ppt	https://javascript.info	ES6 Tutorials
21-22	Callbacks, Promises, Async/Await	CLO01	Book1 Ch9	PPT, Whiteboard	Flipped Classroom	https://github.com/ssagar1999/chitkara_ppt	https://developer.mozilla.org	NPTEL – JavaScript
23-24	Fetch API, REST APIs, JavaScript Mini Project	CLO01	Book1 Ch10	PPT, Live Demo	Project-Based Learning	https://github.com/ssagar1999/chitkara_ppt	https://developer.mozilla.org	Fetch API Tutorials
25-26	Introduction to React, Vite, JSX	CLO02	Book2 Ch1	PPT, Live Coding	Hands-on Practice	https://github.com/ssagar1999/chitkara_ppt	https://react.dev	React Official Videos
27-28	Components, Props, Component Reusability	CLO02	Book2 Ch2	PPT	Collaborative Coding	https://github.com/ssagar1999/chitkara_ppt	https://react.dev	React Components
29-30	State Management using useState(), Event Handling	CLO02	Book2 Ch3	PPT, IDE Demo	Coding Exercise	https://github.com/ssagar1999/chitkara_ppt	https://react.dev	React Hooks
31-32	Forms, Conditional Rendering, Lists & Keys	CLO02	Book2 Ch4	PPT	Pair Programming	https://github.com/ssagar1999/chitkara_ppt	https://react.dev	React Forms
33-34	useEffect(), API Calls, Dependency Array	CLO02	Book2 Ch5	PPT, Live Demo	Case Study	https://github.com/ssagar1999/chitkara_ppt	https://react.dev	React useEffect
35-36	React Router, Route Parameters, Nested Routes	CLO02	Book2 Ch6	PPT	Problem Solving	https://github.com/ssagar1999/chitkara_ppt	https://reactrouter.com	React Router
37-38	Context API, Axios, Global State	CLO02	Book2 Ch7	PPT, Live Coding	Collaborative Learning	https://github.com/ssagar1999/chitkara_ppt	https://axios-http.com	React Context API
39-40	Tailwind	CLO0	Book2	PPT	Hands-on	https://gith	https://tailwindcss	Tailwind

	CSS, Authentication UI	2	Ch8		Practice	ub.com/ssagar1999/chitkara_ppt	.com/docs	CSS
41-42	React Project Development & Best Practices	CLO02	Book2 Ch9	PPT	Project-Based Learning	https://github.com/ssagar1999/chitkara_ppt	https://react.dev	React Project Demo
43-44	Introduction to Node.js, Runtime, Event Loop	CLO03	Book3 Ch1	PPT, Whiteboard	Think-Pair-Share	https://github.com/ssagar1999/chitkara_ppt	https://nodejs.org/docs	NPTEL – Node.js
45-46	Modules, File System, Path Module, Streams	CLO03	Book3 Ch2	PPT, Coding Demo	Hands-on Activity	https://github.com/ssagar1999/chitkara_ppt	https://nodejs.org/api	Node.js Modules
47-48	HTTP Module, NPM, Environment Variables	CLO03	Book3 Ch3	PPT	Coding Exercise	https://github.com/ssagar1999/chitkara_ppt	https://docs.npmjs.com	NPM Tutorials
49-50	Express.js, Routing, Middleware	CLO03	Book3 Ch4	PPT, Live Coding	Problem-Based Learning	https://github.com/ssagar1999/chitkara_ppt	https://expressjs.com	Express.js Tutorials
51-52	REST APIs, CRUD Operations, Validation, Error Handling	CLO03	Book3 Ch5	PPT	API Development	https://github.com/ssagar1999/chitkara_ppt	https://expressjs.com	Postman Learning
53-54	JWT Authentication, bcrypt, Multer, Postman Testing	CLO03	Book3 Ch6	PPT, Live Demo	Hands-on Practice	https://github.com/ssagar1999/chitkara_ppt	https://jwt.io	JWT Tutorials
55-56	MongoDB, MongoDB Atlas, CRUD Operations	CLO04	Book4 Ch1	PPT, Demonstration	Hands-on Lab	https://github.com/ssagar1999/chitkara_ppt	https://www.mongodb.com/docs	MongoDB University
57-58	Mongoose, Schemas, Models, Relationships, Populate	CLO04	Book4 Ch2	PPT, Live Demo	Coding Exercise	https://github.com/ssagar1999/chitkara_ppt	https://mongoosejs.com/docs	MongoDB University
59-60	MERN Stack Integration, Authentication Flow, Protected Routes, Final Project Demonstration & Code Review	CLO05	Book4 Ch3	PPT, Live Demonstration	Capstone Project Presentation	https://github.com/ssagar1999/chitkara_ppt	https://react.dev, https://expressjs.com, https://www.mongodb.com/docs	MERN Stack Project Demo

8. Remedial Classes

After every Sessional Test, different types of learners will be identified, and special discussions will be planned and scheduled accordingly for the slow learners.

Learner Type-I	Learner Type- II	Learner Type- III
Remedial Classes, Doubt Sessions, Guided Tutorials	Workshop, Doubt Session	Coding Competitions, Project

9. Self-Learning

Assignments to promote self-learning, survey of contents from multiple sources.

S. No.	Topics	CLO	ALM	References/MOOCS
1	**Backend Fundamentals:** Introduction to Backend Engineering, Client-Server Architecture, Internet, HTTP/HTTPS, Request-Response Cycle, Node.js Runtime and Installation.	CLO01	Think–Pair–Share	https://nodejs.org/en/learnhttps://developer.mozilla.org/en-US/docs/Web/HTTPhttps://javascript.info
2	**Node.js Programming:** NPM, Core Modules, File System, Events, HTTP Server, Express.js, Middleware, Asynchronous Programming, and REST APIs.	CLO02	Problem-Based Learning / Hands-on Activity	https://expressjs.com/https://learning.postman.com/https://nodejs.org/en/docs
3	**Database Integration and Security:** MongoDB, Mongoose ODM, CRUD Operations, JWT Authentication, Authorization, and API Security.	CLO03	Case Study / Project-Based Learning	https://www.mongodb.com/docs/https://mongoosejs.com/docs/https://jwt.io
4	**Scalable Backend Applications:** Socket.IO, EJS, Git & GitHub, REST API Best Practices, Caching, and Performance Optimization.	CLO04	Collaborative Learning / Mini Project	https://socket.io/docs/https://git-scm.com/docshttps://redis.io/learn
5	**AI Model Integration and Deployment:** FastAPI/Flask, Docker, Kubernetes, Model Serving, Monitoring, and MLOps.	CLO05	Project-Based Learning / Capstone Project	https://fastapi.tiangolo.com/https://docs.docker.com/https://kubernetes.io/docs/https://ml-ops.org

10. Delivery Details of Content Beyond Syllabus

Any content delivered beyond the syllabus will be planned for all students, and the schedule will be notified accordingly.

S. No.	Advanced Topics, Additional Reading, Research papers and any	CLO	POs	ALM	References/MOOCS
1	Modern Backend Development: Advanced Express.js Architecture, REST API Design Best Practices, Middleware Design Patterns, API Documentation (Swagger/OpenAPI), and Backend Performance Optimization.	CLO01, CLO02	PO1, PO2, PO3, PO5, PO11	Think–Pair–Share	https://expressjs.com/https://swagger.io/docs/https://roadmap.sh/backend
2	Scalable Backend Systems: MongoDB Performance Tuning, Database Indexing, Redis Caching, Load Balancing, Rate Limiting, API Security, and Authentication using JWT and OAuth.	CLO03, CLO04	PO1, PO2, PO3, PO5, PO6, PO11	Problem-Based Learning	https://www.mongodb.com/docs/https://redis.io/learnhttps://jwt.io/https://auth0.com/docs
3	Cloud-Native Backend and AI Integration: Docker, Kubernetes, FastAPI/Flask, MLOps, Model Deployment, Model Monitoring, CI/CD Pipelines, and AI-Powered Backend Services.	CLO04, CLO05	PO1, PO3, PO5, PO7, PO10, PO11	Case-Based Learning	https://fastapi.tiangolo.com/https://docs.docker.com/https://kubernetes.io/docs/home/https://ml-ops.org

11. Evaluation Scheme & Components

Assessment Type	Evaluation Component	Type of Component	No. of Assessments	% Weightage of Component	Max. Marks	Mode of Assessment	CLO
Formative	Component1	Continuous Assessments	02*	50%	50	Online	CLO01 - CLO05
Formative	Component 2	Final Assessment	01**	50%	50	Online	CLO01 - CLO05
Total			100%				

* 02 continuous assessments (each of 25 marks) will be taken for the evaluation as final marks.
**01 Final evaluation will be considered for evaluating the entire/complete project and to be eligible to appear for the final project evaluation, attendance must be at least 75%.

12. Syllabus of the Course

Subject: Back End Engineering		**Subject Code:** 24CAI0303	
S. No.	**Topic (s)**	**No. of Lectures**	**Weightage %**
1	**Backend Fundamentals:** Introduction to Backend Engineering, Client-Server Architecture, Internet, Data	9	15%

	Transfer, HTTP & HTTPS, Node.js Architecture, Installation & Setup, NPM, Package Management, Node.js Core Modules (Path, OS, File System, Events).		
2	**Node.js & Express.js Development:** HTTP & URL Modules, Creating Local Servers, Import & Export Modules, Debugging, Nodemon, Package Management, Express.js, Routing, Middleware, Request Lifecycle, Error Handling, Asynchronous Programming, REST API Development and Testing using Postman.	18	30%
3	**Database Design & Backend Security:** Database Fundamentals, SQL vs NoSQL, MongoDB, CRUD Operations, Mongoose ODM, Authentication & Authorization, JWT Authentication, Session Management, API Security.	15	25%
4	**Advanced Backend Development:** Socket.IO, Real-Time Applications, EJS Templating, Git & GitHub, REST API Best Practices, API Documentation, Pagination, Filtering, Rate Limiting, Caching, Performance Optimization using Redis.	12	20%
5	**AI Model Integration & Deployment:** FastAPI/Flask, AI Model Serving, Docker, Kubernetes, Model Deployment, Monitoring, Versioning, MLOps Fundamentals	6	10%

13. Academic Integrity Policy

Education at Chitkara University builds on the principle that excellence requires freedom where Honesty and integrity are its prerequisites. Academic honesty in the advancement of knowledge requires that all students and Faculty respect the integrity of one another's work and recognize the importance of acknowledging and safeguarding intellectual property. Any breach of the same will be tantamount to severe academic penalties.

This Document is approved by

Designation	Name	Signature
Course Coordinator	Dr. Palak Goyal	
Head Academic Delivery	Dr. Vandana Sood	
Dean (CSE-AI)	Dr. Sushil Kumar Narang	
Date (DD/MM/YYYY)	23/06/2026	